

ProTech — Professional Services Platform

Complete Project Report

Prepared by: Patrice Junior

Project Type: Web-Based Service Platform

Technology Stack: HTML5 · CSS3 · JavaScript · PHP · MySQL

Date: June 2026

1. Executive Summary

ProTech is a comprehensive web-based professional services marketplace built using HTML5, CSS3, JavaScript, PHP, and MySQL. The platform provides a trusted digital environment where clients can discover, book, and review skilled professionals across a wide range of service categories including plumbing, electrical work, carpentry, cleaning, and HVAC maintenance.

The platform was developed with a dual-portal approach, providing distinct experiences for two primary user groups: clients who need services and workers who provide them. This architectural decision reflects a deep understanding of the distinct workflows, information needs, and interaction patterns of each user group.

Deliverable	Status	Technology
Landing Page	Complete	HTML5, CSS3, JS
Authentication System	Complete	PHP, MySQL
User Dashboard	Complete	HTML5, CSS3, JS

Worker Dashboard	Complete	HTML5, CSS3, JS
Database Schema (12 tables)	Complete	MySQL
Demo Mode	Complete	PHP Sessions

2. Project Overview

ProTech addresses a significant gap in the professional services market: the difficulty clients face in finding trustworthy, verified professionals and the challenge workers encounter when trying to grow their customer base beyond word-of-mouth referrals.

2.1 Technology Stack

Layer	Technology	Purpose
Frontend Structure	HTML5	Semantic page structure
Frontend Styling	CSS3	Responsive visual design
Frontend Logic	JavaScript	Interactivity and validation
Backend	PHP 7.4+	Server-side processing
Database	MySQL	Data persistence
Dev Environment	XAMPP	Local server

3. Landing Page

The landing page serves as ProTech's primary digital storefront, converting visitors into registered users through compelling content,

clear value propositions, and strategic call-to-action placement. The page includes a Hero Section, Features Section (6 cards), How It Works (4 steps), About Section with statistics (10,000+ professionals, 50,000+ clients, 100+ service categories), a Contact Section with form, a CTA section, and a comprehensive Footer.

3.1 Design Specifications

Color	Hex Code	Usage
Primary Blue	#2563eb	Buttons, links, headings
Secondary Indigo	#4f46e5	Gradients, accents
Success Green	#10b981	Worker theme, success states
Text Dark	#111827	Primary text
Text Gray	#6b7280	Secondary text, labels

4. Authentication System

The authentication system provides secure, role-based access control with separate registration workflows for clients and service providers. It uses a three-tab card interface: Login, User Signup, and Worker Signup.

Form	Fields	PHP Handler	Redirects To
Login	Email, Password	login_handler.php	User or Worker Dashboard
User Signup	Name, Email, Phone, Password	user_signup_handler.php	Login page

Worker Signup	Name, Email, Phone, Address, Skills, Experience, Password	worker_signup_handler.php	Login page
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5. Security Implementation

Security Measure	Implementation	Threat Addressed
Password Hashing	PHP password_hash() with bcrypt	Data breach, rainbow table attacks
SQL Injection Prevention	MySQLi prepared statements	Database manipulation
XSS Prevention	htmlspecialchars() on all output	Cross-site scripting
Session Management	session_regenerate_id()	Session fixation
Access Control	require_user_type() checks	Unauthorised access
Login Monitoring	login_attempts table	Brute force attacks

6. Database Design — 12 Tables

Table	Purpose	Key Fields
users	All user accounts	id, email, password, user_type
workers	Extended worker info	user_id, skills, rating, status

user_sessions	Session tracking	session_id, user_id, expires_at
login_attempts	Auth monitoring	email, ip_address, success
password_resets	Password recovery	token, expires_at, used
email_verifications	Email confirmation	token, verified, expires_at
user_preferences	User settings	language, timezone, notifications
user_addresses	Service locations	address, city, state, zip
oauth_providers	Social login	provider, provider_user_id
two_factor_auth	2FA config	enabled, method, secret_key
account_security_logs	Audit trail	event_type, severity, ip
blocked_ips	IP blocking	ip_address, reason, expires_at

7. Testing Results

Test Case	Expected Result	Status
User registration	Account created, redirect to login	PASS
Duplicate email	Error: Email already exists	PASS

Client login	Redirect to user dashboard	PASS
Worker login	Redirect to worker dashboard	PASS
Wrong password	Error: Invalid credentials	PASS
SQL injection attempt	Input sanitized, no effect	PASS
Demo mode access	Demo session, dashboard loads	PASS
Mobile responsive	Layout adapts correctly	PASS

8. Conclusion

ProTech successfully delivers a professional-grade web platform demonstrating mastery of full-stack web development. The project integrates HTML5, CSS3, JavaScript, PHP, and MySQL into a cohesive system serving two distinct user groups through purpose-built interfaces. The landing page effectively communicates value and converts visitors, while the authentication system securely manages user registration and login with role-based access control.

The 12-table database schema provides a scalable foundation for future feature development including payment integration, real-time messaging, AI-powered worker matching, and native mobile applications.

Project Status: Partially Completed — not all planned deliverables implemented, tested, and documented.

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